Cover page with signatures:

Project Team Charter

MIME 497: Capstone I

Signature Cover Page

Each team member will copy the following statement in their own handwriting (LEGIBLY) in one of the designated areas below:

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

Print Name: Andrew Dillon_____ **Signature:** Andrew Dillon

Handwritten Statement:

I. Agree to do an equal amount of work in the Team. I understand that my grade will reflect my effort in the team

Print Name: **Aiden Aerni** **Signature:** Aiden Aerni

Handwritten Statement:

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort on the team.

Print Name:Joshua Calderon_____ **Signature:**_____

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team

Joshua Calderon

Team purpose:

Our purpose for this project is to showcase all our relevant skills to working in the engineering field. Over the winter and spring term, our team of 3 will work to create a project for the school which shows our capabilities to the school, and future employers. To do this, we will create an R/C controlled robot to compete in the school's SUMOBot competition.

The SUMOBot competition is one which consists of multiple teams creating their own version of a robot that is able to push its opponents outside the competition ring. There are set guidelines and standards which each team must adhere to. We will need to design our robot using multiple engineering disciplines such as: Robotics, Programming, Mechanical Design, Material Selection, Manufacturing, and Physics.

Team Goals:

Our main team goal is to create a working robot that may compete amongst the other robots while adhering to the official rules. Our goal is to get an A in the both Capstone classes with a reach goal being our robot having a positive win ratio. We also have some sub goals, which include creating organized documentation of our project, becoming familiar with all required disciplines for this project, having a deep understanding of the creation of our project so that we may explain it to other professionals. We aim to treat this Capstone project as a way of presenting our skills to employers in the future, as this is the last large project which is required for our curriculum.

Team Member Personalities and roles:

Andrew Dillon: My personality is someone who prefers to be given orders rather than be the person who gives them while also making sure that everything on my end is running smoothly and helping whenever issues may arise. However, for this project, I am going to be checking deadlines and making sure that we are not falling behind on assignments for that week. For non-management sections of this project, I will be overseeing the mechanical aspect of the designing process while working with Joshua Calderon on the electrical/coding aspects of the robot.

Joshua Calderon: I would describe my team personality as someone who tries to lead and provide support when needed from other team members. I think communication is one of my greatest strengths as I always try to respond quickly, giving updates regarding my task. Tentatively, until we get a better scope of the electrical/coding side of the project, my role in Sumo Bot is to be responsible for the coding and electrical components of the robot.

Aiden Aerni: I have a creative personality and like to bring new ideas to the project. I like to find ways to solve problems efficiently and find fulfillment in creating a good product. My team role has been decided to be more focused on mechanical design. I intend to use CAD software such

as NX, Solidworks, or Onshape in this role. This is subject to change and of course interjection from the other members of the group as we will all have design preferences and ideas. It is still early in our project so our roles and workload are likely subject to change.

Ground Rules:

Main meeting times will occur Monday from 6-8pm after the Capstone hybrid lecture in order to retain the information from class and immediately apply it into our work for the week. However we will have some extra sub meetings if we could not fit all we needed to do just in that single meeting, preferably about 2-3 quicker submeetings throughout the week from 4-5 or 5-6 most weekdays Alongside, we will be attempting to have weekly update meetings with our advisor to guide us on possible next steps we should take in the design process. Our meetings should have discussion topics agreed upon before the meeting such that we can reduce time wasting from not communicating clearly. In issues that are not a unanimous decision we will attempt to discuss to see the best way forward, but if a unanimous decision is not possible we will go with majority vote.

We expect everyone to give their best effort regardless of how good it is as there is always something that we can improve upon. But in the meetings, we will have a short run through on what we accomplished that week, so there would be accountability from everyone if you could not reasonably describe what you did or did not do that week.

Depending on what team assignments are due that week, we may delegate members who have signified that they have finished their required sections early to work jointly with additional team members in order to finish the most extensive projects. In order to reduce confusion about what, team members may only be assigned 1 or 2 subsections to begin with and when they signified they have finished, they would be assigned new subsections to collaborate on.

Potential barriers and coping strategies:

Team Barriers:

- Potential barriers for effective teamwork may result from poor communication between team members, one way to solve that is by having members remain in daily communication while remaining professional about reminders and to take the advice in willingly by trying to have each member not need multiple reminders before fixing the issue.
- Another issue that could come up is individual members feeling left out of some sections of the project as they feel their help is not wanted. A possible solution is by having each member contribute to each section is a small way could help them feel that their work is valued

Sponsor Barriers:

- Sponsor meetings could be scheduled at improper times if there is not a time that works for everyone. But by offering many different time slots, we can solve this problem and show our sponsor that we are able to solve their issues and
- The sponsor may ask for something that we are not realistically able to complete or may switch their demand in a meeting and may blame a resulting issue on us, by taking detailed meeting notes when meeting with our sponsor, we can hope to avoid this problem before it becomes something major, while also providing accountability to the team members as well as the sponsor.